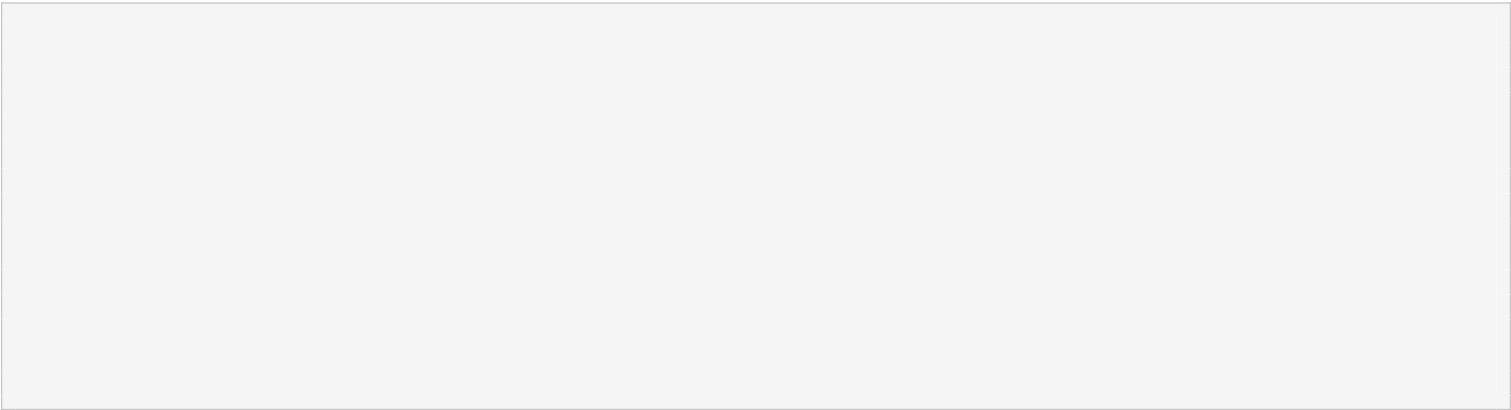
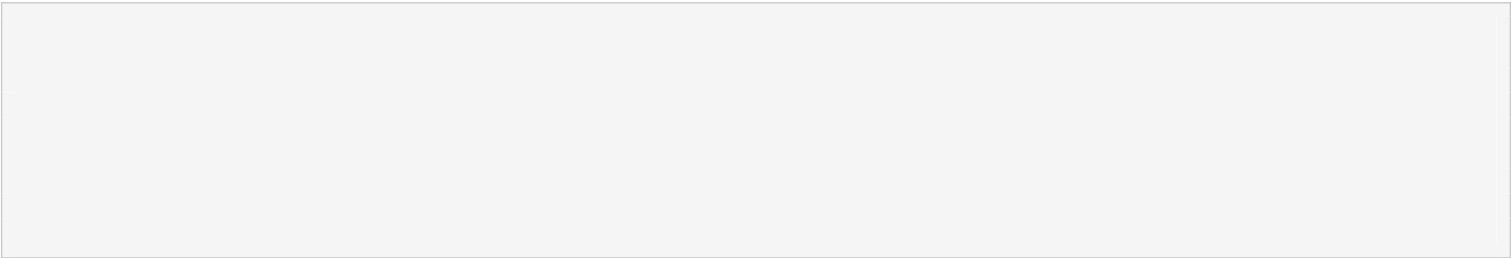
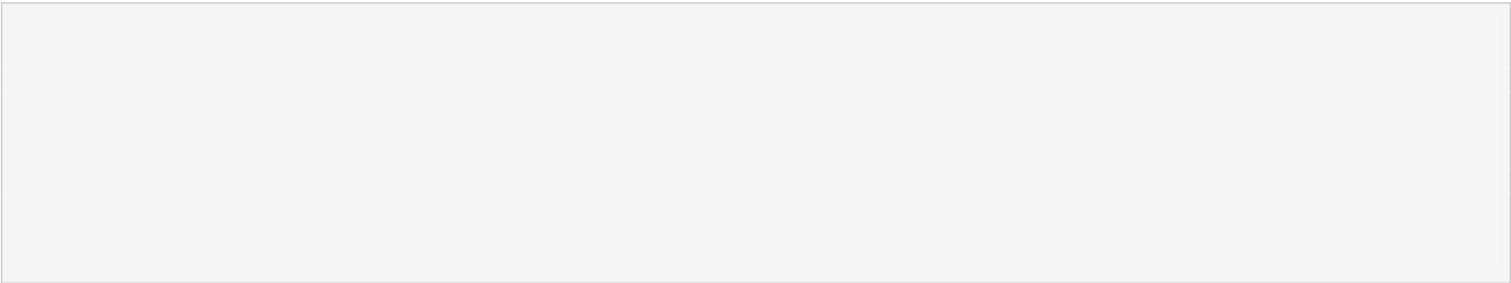


$(2, 1, 1, 0)$



$(2, 1, 1, 0)$



(k, k, k, k)

$$\|/4\\(2,1,1,0)$$

$$(0,0,0,0)(1,0,0,0)(0,1,0,0)(0,0,1,0)\frac{1}{4}$$

$$\frac{P_0,\ldots,P_3}{VS3=\sqrt{9/8}}$$

$$\mid_{5\times 5}\mid_{3\times 3}\qquad\qquad\qquad\mid\mid/4$$

$$\frac{1}{4}$$

$$(P_1-P_0)$$

$$d_{ij}P_iP_j$$

$$V_{ivm}=S3\cdot V_{xyz}S3=\sqrt{9/8}$$

$$V_{ivm}=S3\cdot V_{xyz}S3=\sqrt{9/8}$$

$$P_1-P_0P_2-P_0P_3-P_0\frac{1}{4}$$

$$[i,j]=\mathbb{E}[\partial_{\theta_{\boldsymbol{i}}} \; p(x;\theta) \; \partial_{\theta_{\boldsymbol{j}}} \; p(x;\theta) \,]$$

$$_{288}V^2=\begin{pmatrix}0&1&1&1&1\\1&0&d_{01}^2&d_{02}^2&d_{03}^2\\1&d_{10}^2&0&d_{12}^2&d_{13}^2\\1&d_{20}^2&d_{21}^2&0&d_{23}^2\\1&d_{30}^2&d_{31}^2&d_{32}^2&0\end{pmatrix}$$

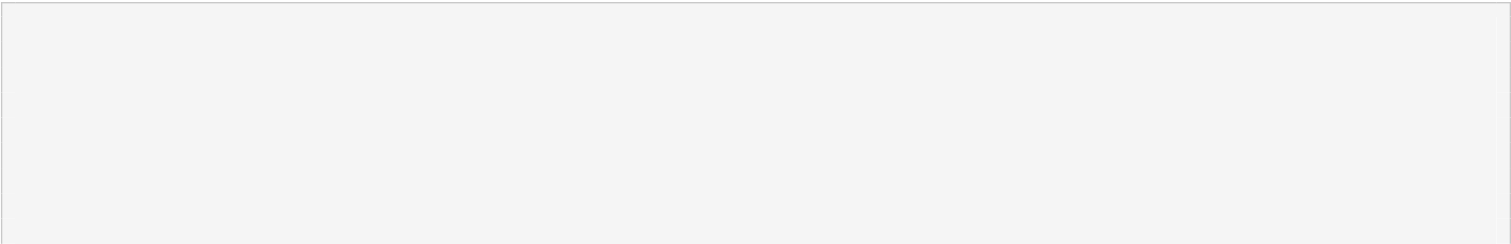
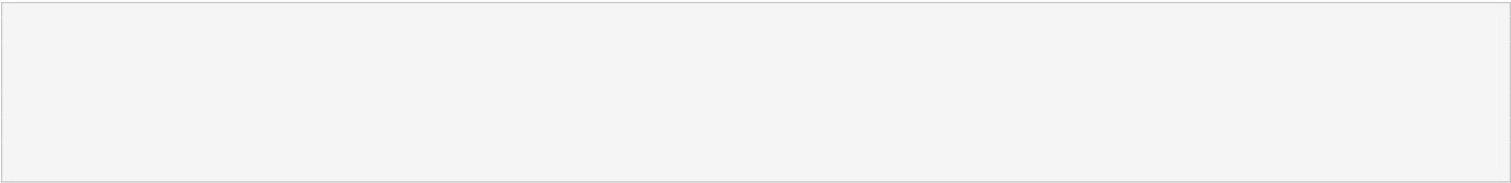
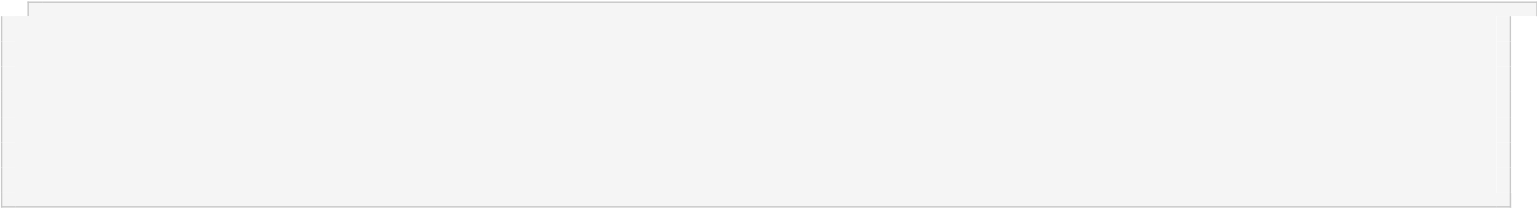
$$S3=\sqrt{9/8}$$

$$a,b,cd,e,f$$

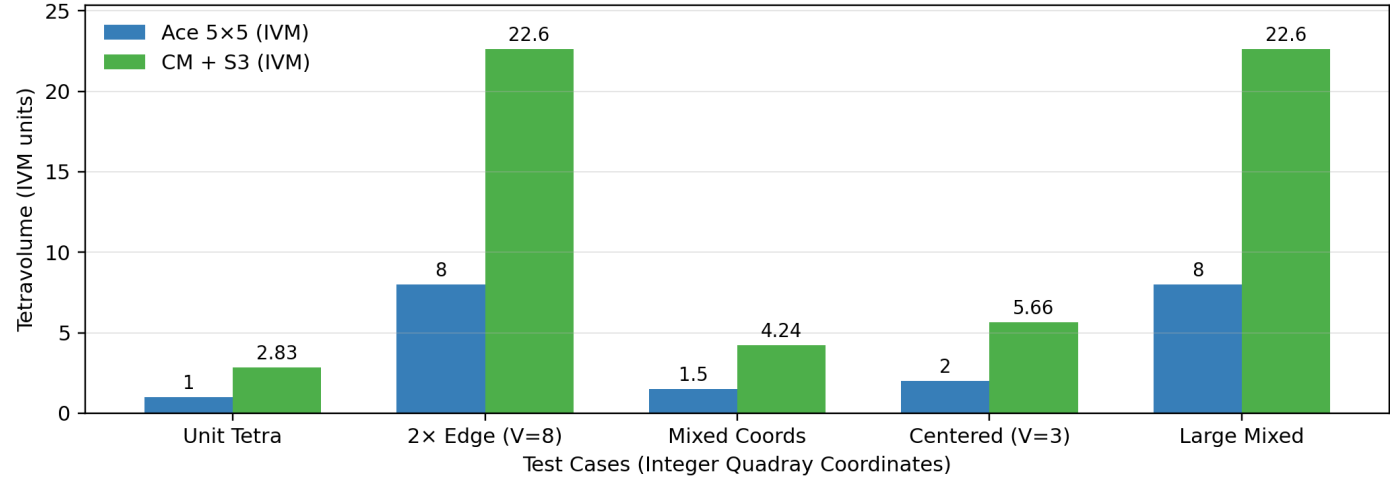
$$V_{ivm}=S3\cdot V_{xyz}S3=\sqrt{9/8}$$

$$P_1-P_0P_2-P_0P_3-P_0\frac{1}{4}$$

$$S3=\sqrt{9/8}$$



Validation: Bridging vs Native Tetravolume Formulations

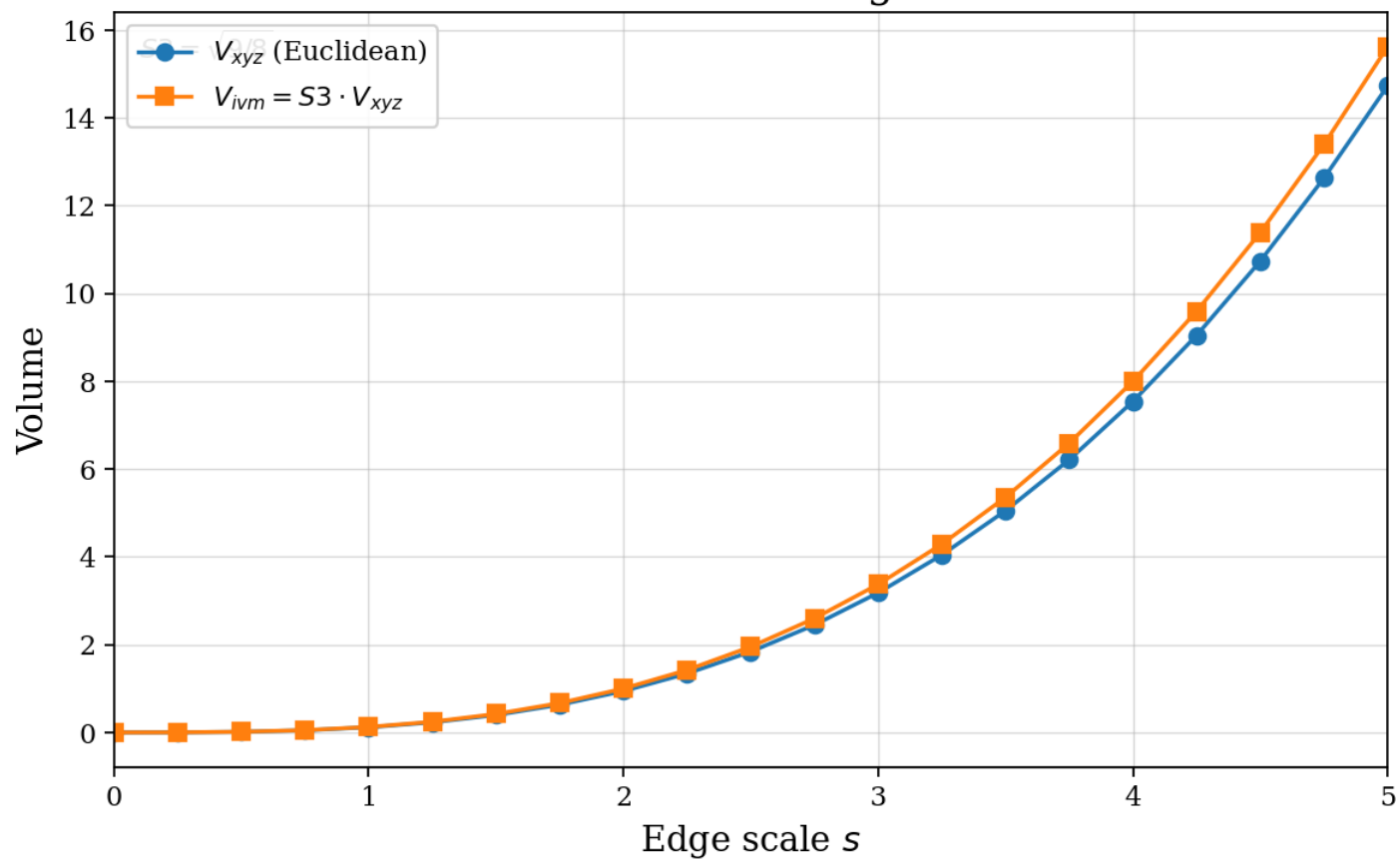


Test cases: Unit tetra (V=1), 2x edge scaling (V=8), mixed coordinates, centered (V=3), large mixed.
Both methods agree at machine precision, validating the S3 conversion factor. Data: quadmath/output/bridging_vs_native.csv

$S3 = \sqrt{9/8}$

(2, 1, 1, 0)

Tetra volume vs edge scale



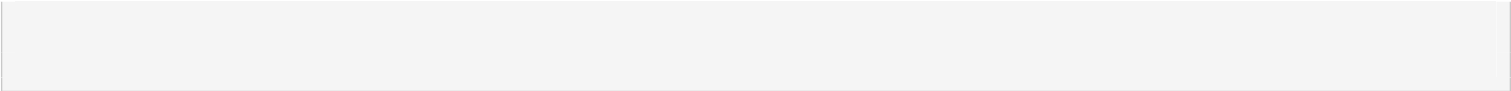
$$V = \frac{\sqrt{2}}{12} \cdot L^3 V_{ivm} = \frac{1}{8} \cdot L^3 S3 = \sqrt{9/8} \approx 1.06066$$

A 3D plot showing various polyhedra with their volumes labeled. The polyhedra include a Tetrahedron (V=1), Cube (V=3), Octahedron (V=4), Rhombic Dodecahedron (V=6), Cuboctahedron (V=20), and Truncated Octahedron (V=20). The plot is set in a 3D coordinate system with axes X, Y, and Z.

The diagram illustrates the relationships between various Platonic solids and their scaled or transformed versions. The nodes and their connections are as follows:

- Tetrahedron (Unit)** ($V=1$)
 - Connects to **Cube** via $\times 3$
 - Connects to **Tetrahedron (2x scale)** via $\times 8$
 - Connects to **Cuboctahedron (Vector Eq.)** via **dual**
- Cube** ($V=3$)
 - Connects to **Octahedron** via $+1$
 - Connects to **Cuboctahedron (Vector Eq.)** via **edge-union**
 - Connects to **Truncated Octahedron** via **shell of 12**
 - Connects to **Cube (2x scale)** via $\times 8$
- Octahedron** ($V=4$)
 - Connects to **Truncated Octahedron** via **truncation**
 - Connects to **Truncated Octahedron** via **Voronoi cell**
 - Connects to **Cube (2x scale)** via **dual**
- Rhombic Dodecahedron** ($V=6$)
 - Connects to **Truncated Octahedron** via $+2$
 - Connects to **Rhombic Dodecahedron (2x scale)** via **dual** and $\times 8$
- Cuboctahedron (Vector Eq.)** ($V=20$)
 - Connects to **Truncated Octahedron** via **dual**
 - Connects to **Tetrahedron (2x scale)** via **dual**
- Truncated Octahedron** ($V=20$)
 - Connects to **Cube (2x scale)** via **dual**
- Tetrahedron (2x scale)** ($V=8$)
- Octahedron (2x scale)** ($V=32$)
- Cube (2x scale)** ($V=24$)
- Rhombic Dodecahedron (2x scale)** ($V=48$)

Volume relationships in IVM tetra-units. Regular tetrahedron ($V=1$) is the fundamental unit.
Scaling relationships: $2 \times$ edge length $\rightarrow 8 \times$ volume ($V \propto L^3$).
Geometric constructions: edge-union, truncation, dual relationships, and Voronoi cells.
All polyhedra constructed with consistent edge lengths and integer Quadray coordinates.



$A, B, CABC$
 $(P_i - P_0)$





